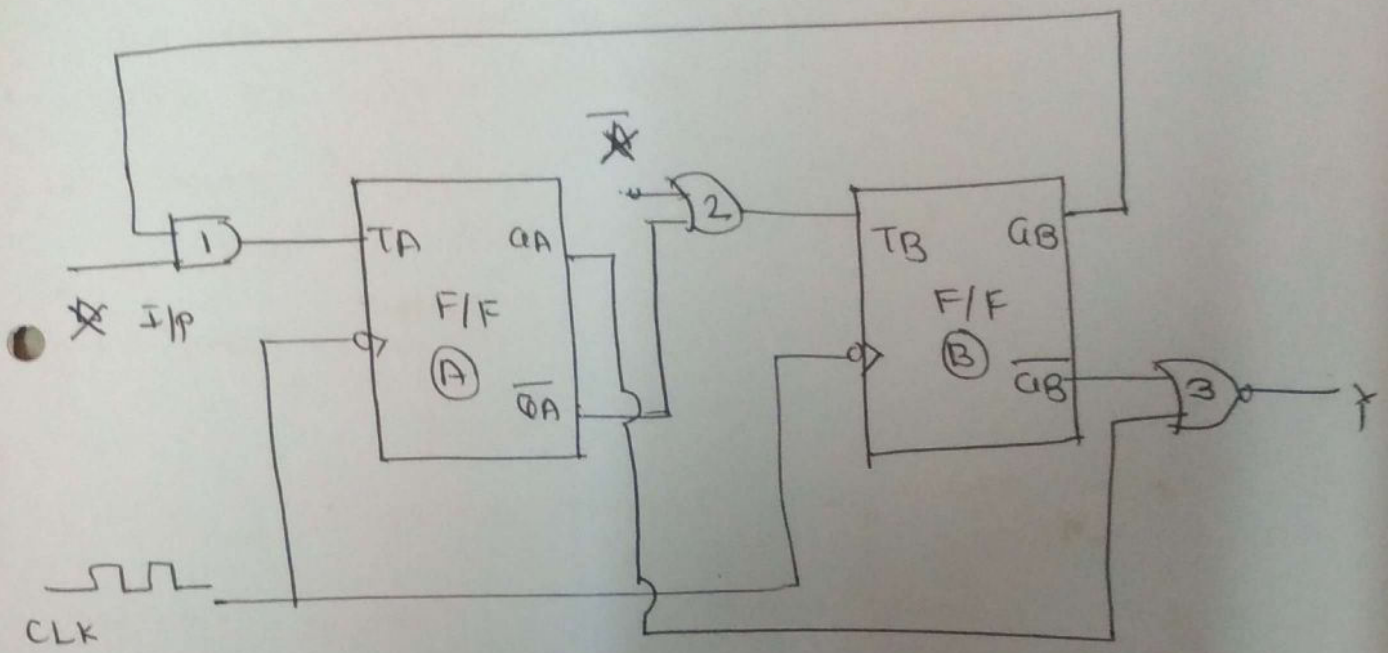
Moore circuit:-

o/p depends on present state of memory element

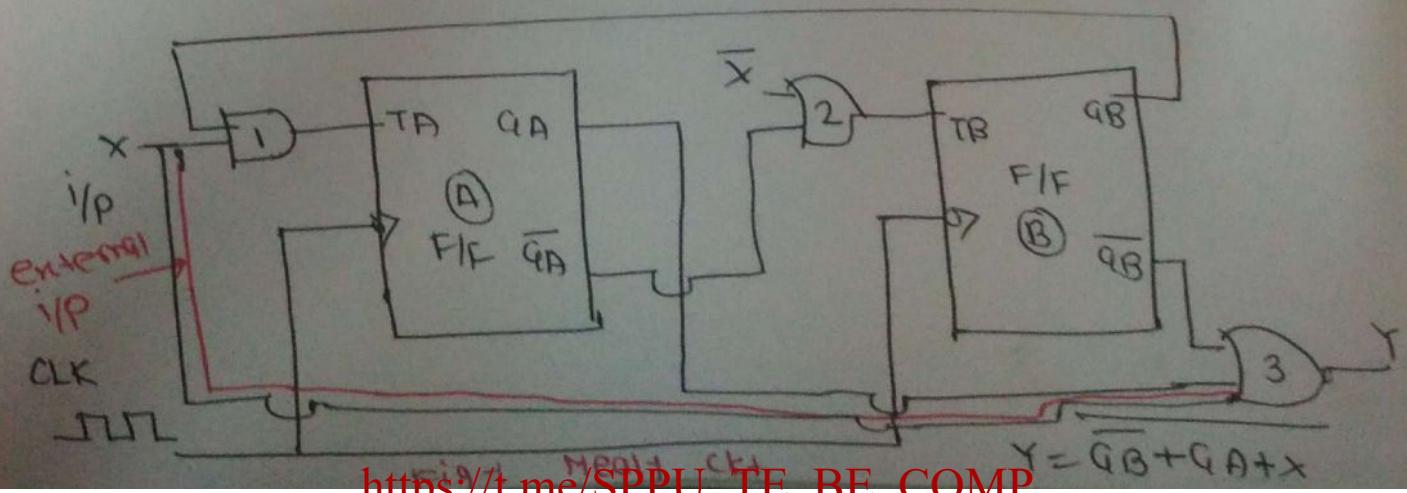


$$Y = \overline{Q_A + \overline{Q_B}}$$

Fig:- Example of Moore ckt.

Mealy ckt.:-

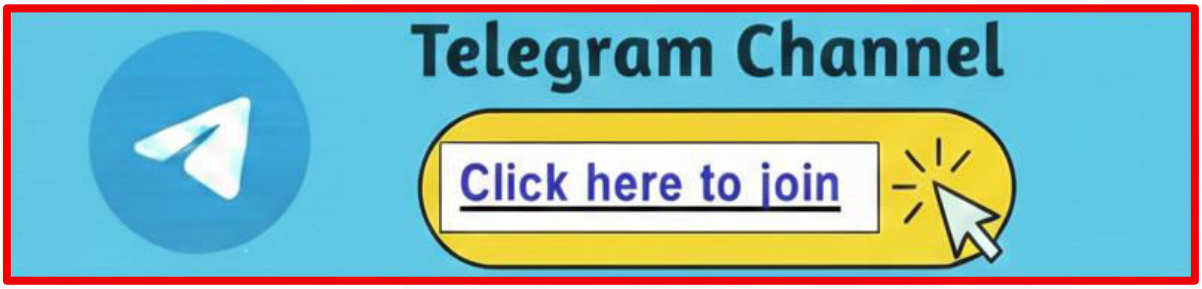
o/p depends on present state of memory element as well as external i/p.

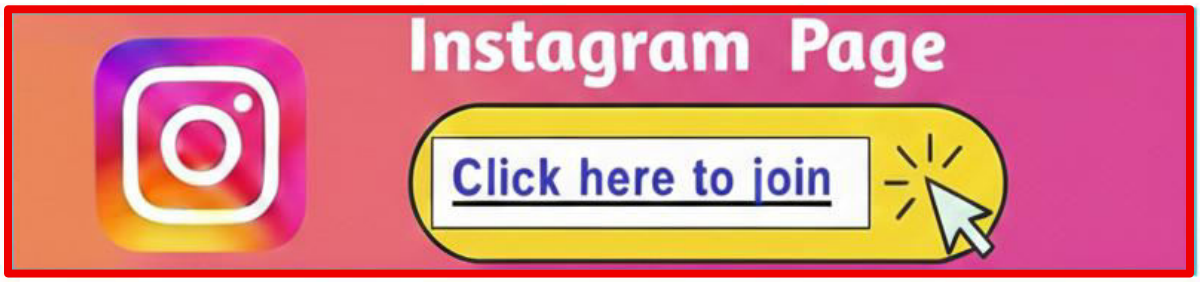


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Comparison of Moore and Mealy Models

Moore Model	Mealy Model
• The final output depends only on the present state of memory elements	• The final output depends on the present state of memory elements and the external inputs
• Synchronous circuit	• Synchronous circuit
• The output changes only after the active clock edge.	• Output can change in between the clock edges if the external inputs changes.
• Output is placed on states	• Output is placed on Transition.
• The implementation of a logic function needs more number of states than the Mealy circuit.	• Implementation of the same logic function requires less number of states than Moore circuit.

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State machine Notation :-

i) I/p variable

All the variable which gets generated outside the sequential circuit is called as input variables

ii) O/p variable :-

All the variable that leave sequential circuit are called as output variable

iii) State variable :-

O/p of Flip-flop is called state variable.

iv) State variable

$$Y = 2^x$$

Y = no. of states x = state variable

Present state :-

Status of all variable at any instant 't'

Next state :-

Status of state variable at instant 't+1'

State Table :-

The relation between the input & output for a sequential circuit.

X = i/p

Y = o/p.

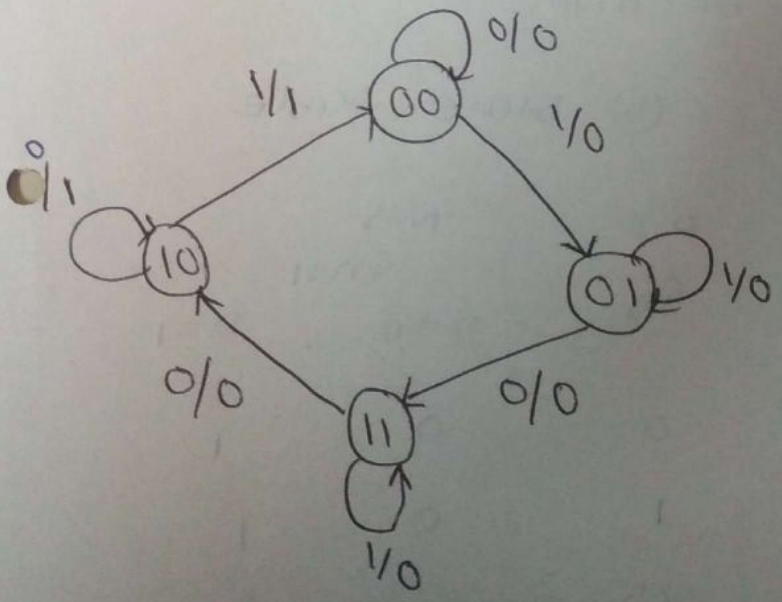
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Present State		Next State				O/P Y	
Q_A	Q_B	$x=0$		$x=1$		$x=0$	$x=1$
		Q_A	Q_B	Q_A	Q_B		
0	0	0	0	0	1	0	0
0	1	1	1	0	1	0	0
1	0	1	0	0	0	1	1
1	1	1	0	1	1	0	0

Fig:- General format state table

State diagram:- (7-7)



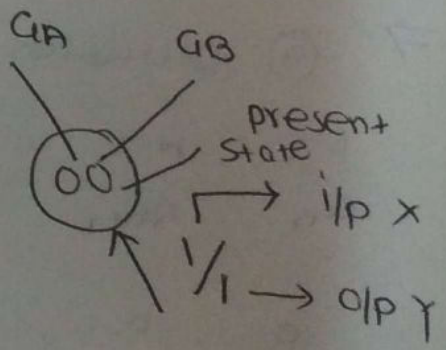
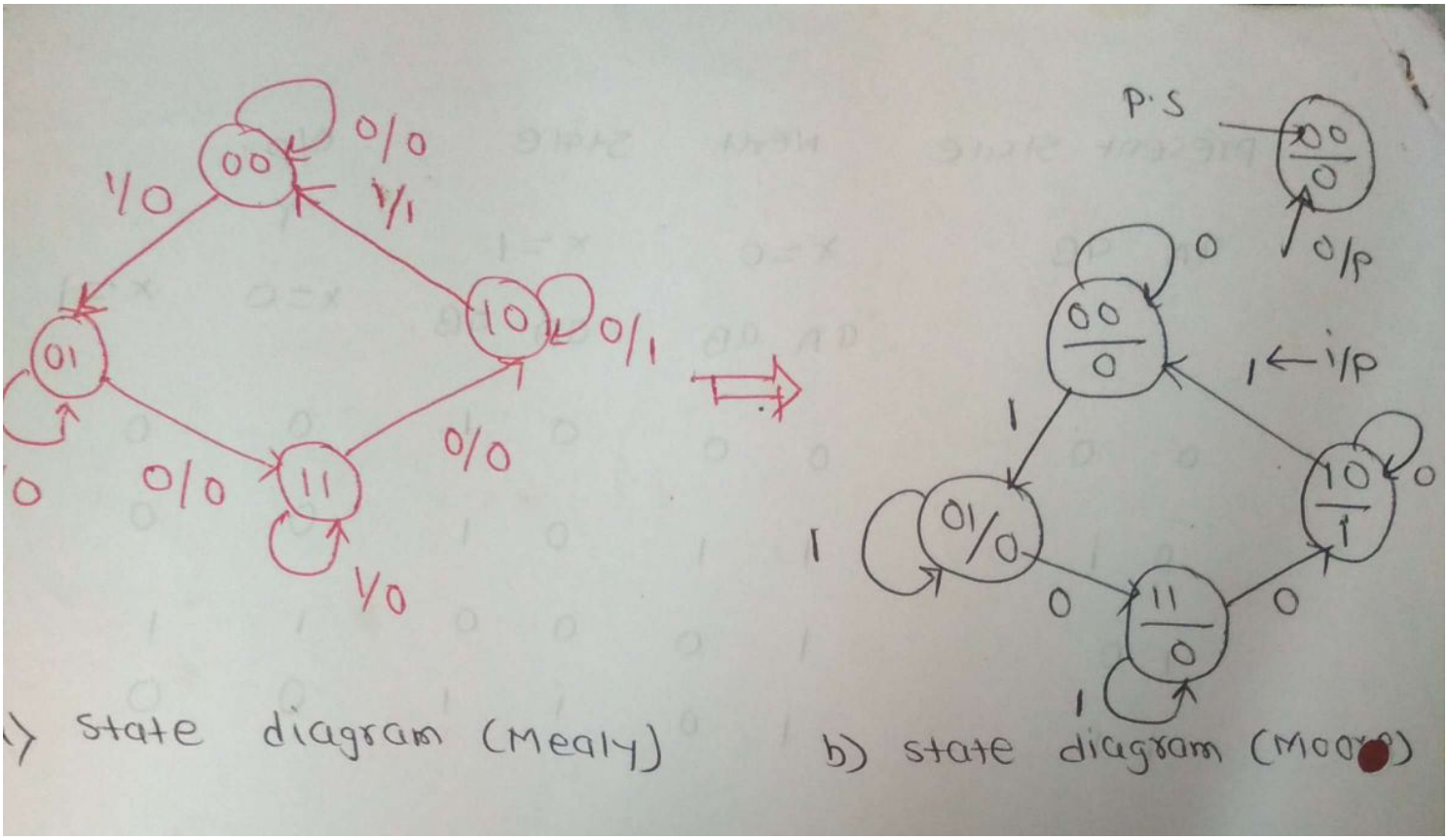


Fig:- state diagram (Mealy)

Information available in state table represented graphically is called state diagram.

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Algorithmic State Machine :-

i) An ASM chart is similar to conventional flowchart but the interprete in different manner

ii) Asm chart is a basically flowchart which represents a hardware algorithm. It is wefyl to design the hardware ~~algorithm~~ of a sequential circuit as per specifications.

ASM chart Notation :-

i) The state box
ii) Decision box
iii) conditional box.

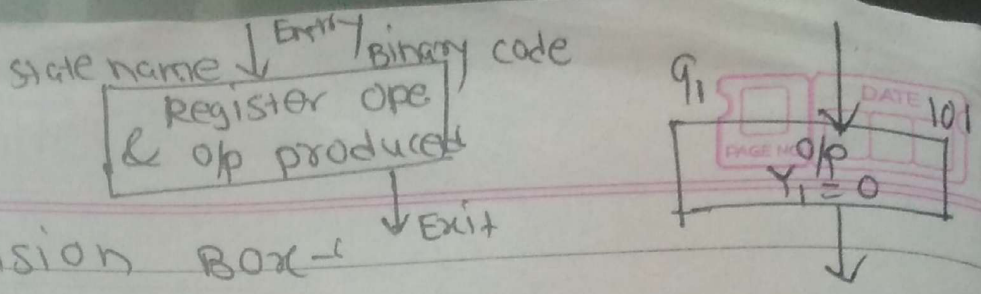
I) State box :-

i) State box is rectangular in shape. Inside this box we write information such as register operation or output signal

ii) one entry & one exit path.

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II) Decision Box :-

Decision box is a dimond shaped box. It has two or more exit path there is only one exit path for the state box.

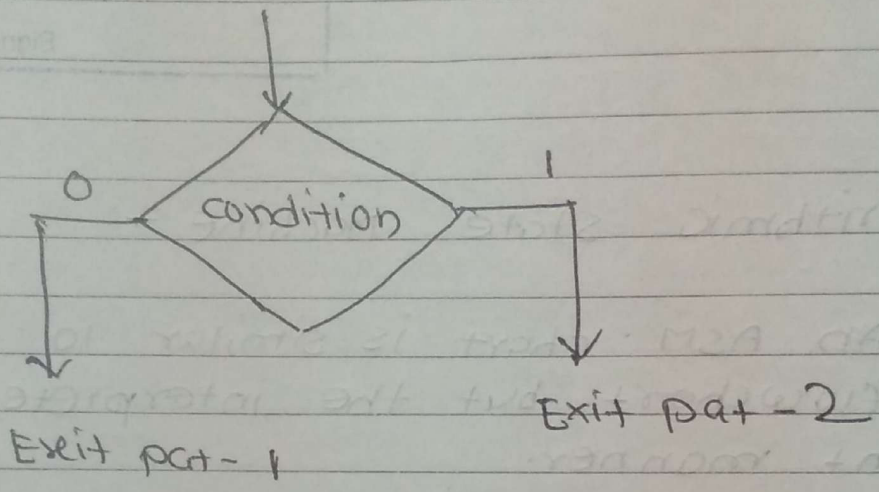
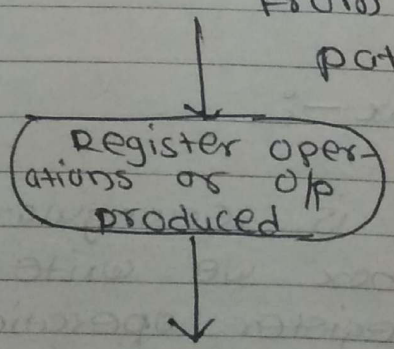


Fig:- Decision box

III) Conditional Box :-

Conditional box is oval in shape. The input path to the conditional box always comes from one of the exit path of decision box.



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Algorithmic state m/c

ASM chart :- (Arithmatic)

ASM chart is special type of flowchart which is suitable for describing the operation of a digital system. sequential

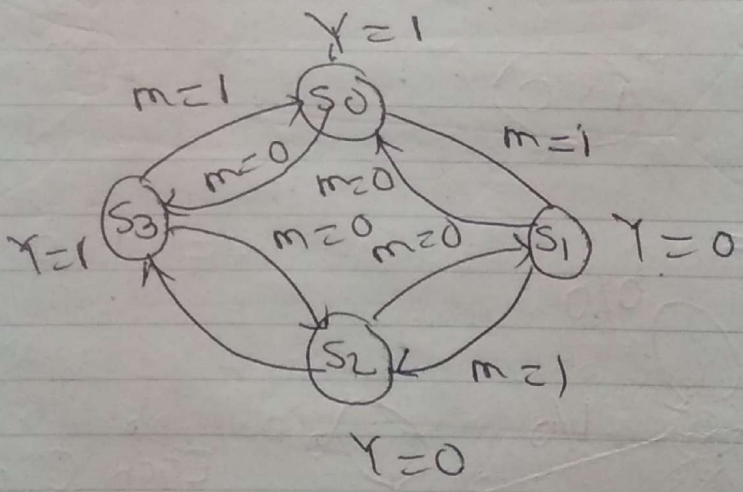
EX ①

Draw an ASM chart and state table of a 2-bit up-down counter having a mode control i/p

⇒

M = 1 → up counting
M = 0 → down counting.

$S_0 \rightarrow 00$
$S_1 \rightarrow 01$
$S_2 \rightarrow 10$
$S_3 \rightarrow 11$

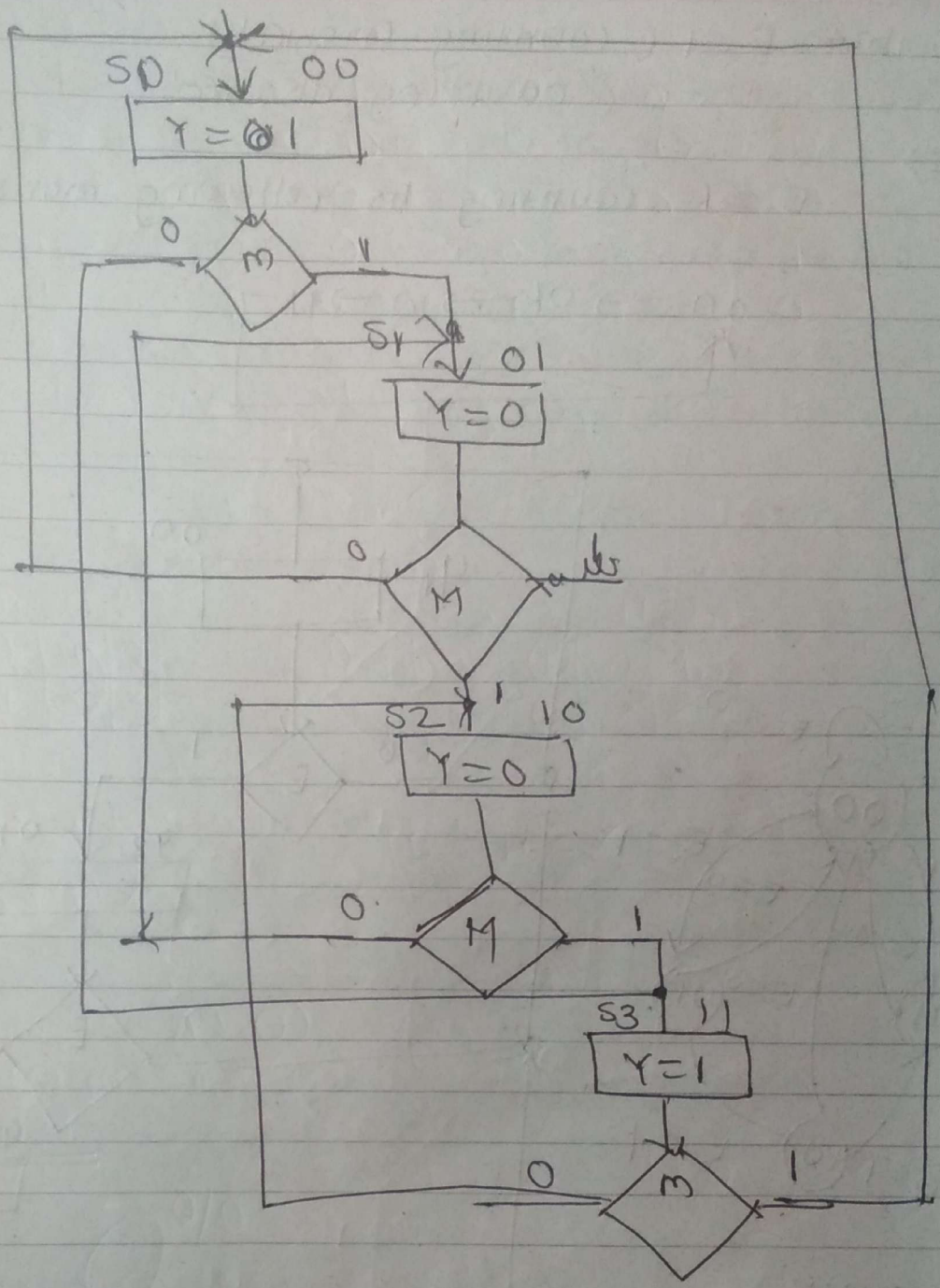


CKt. should produce o/p = 1 when count becomes minimum (00) or maximum (11)

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ASM Chart



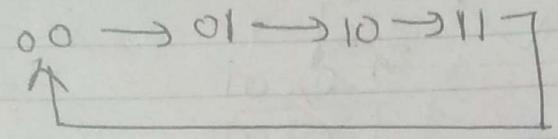
ASM chart for 2-bit up/down counter

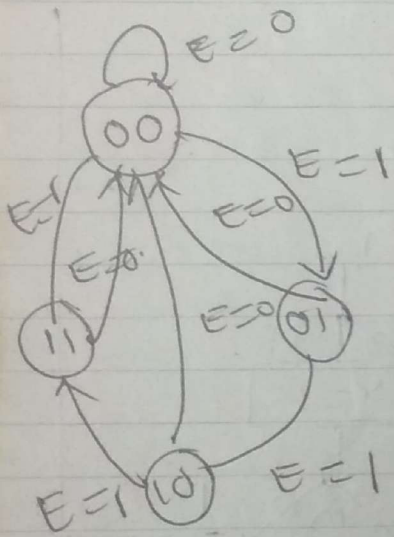
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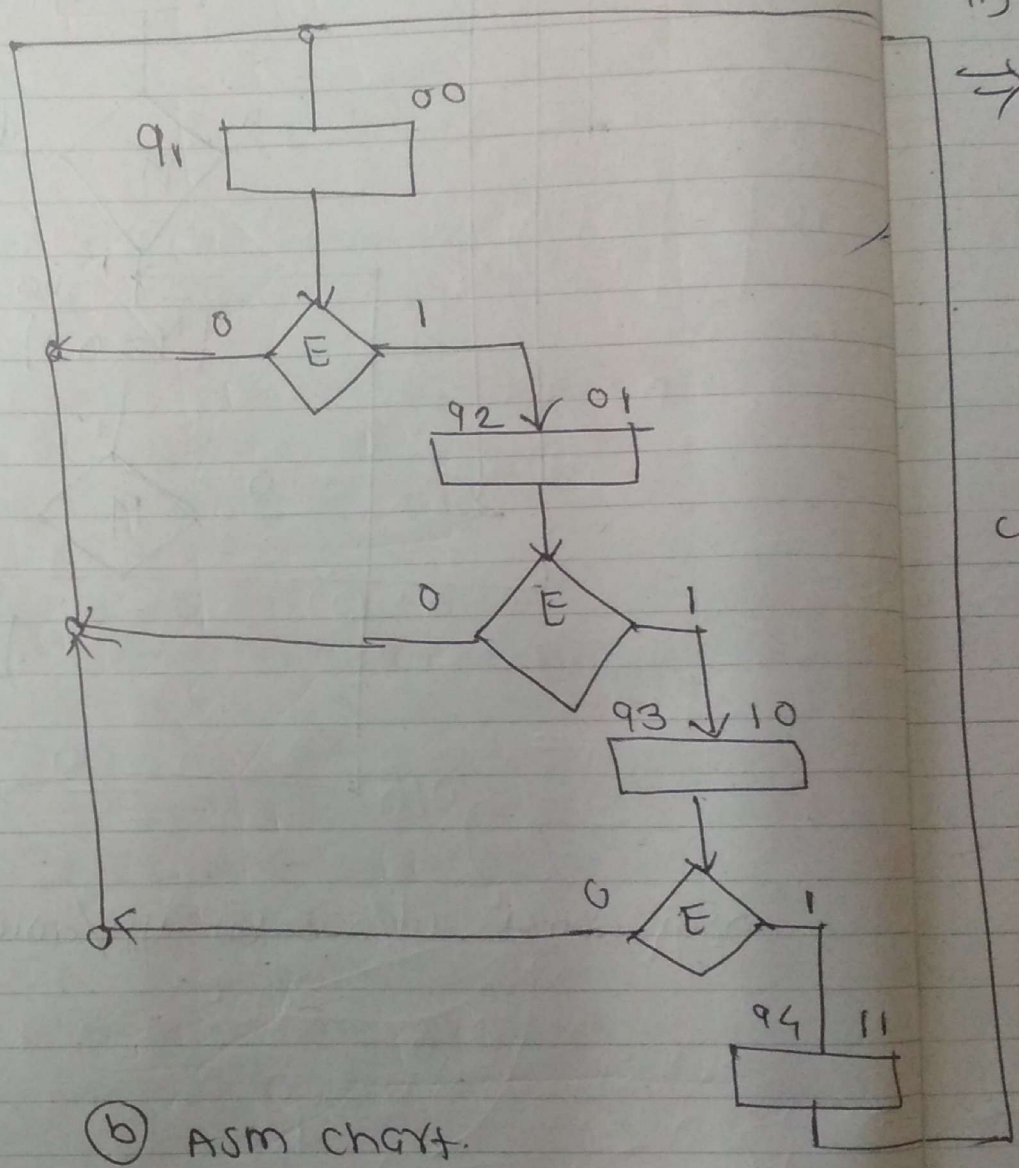
② Draw the ASM chart for a 2-bit binary counter having one enable line E such that E=1 (counting enabled)
E=0 (counting disabled)

⇒ E=1 counting in following manner.





(a) State diagram



(b) ASM chart.

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(3) Draw the ASM chart & state diagram for the synchronous ckt having the following description:-

The circuit has control i/p C, clock & o/p X, Y, Z.

i) If C=1, on every rising edge of the clock code on o/p X, Y & Z changes from 000 → 010 → 100 → 110 → 000 & repeats.

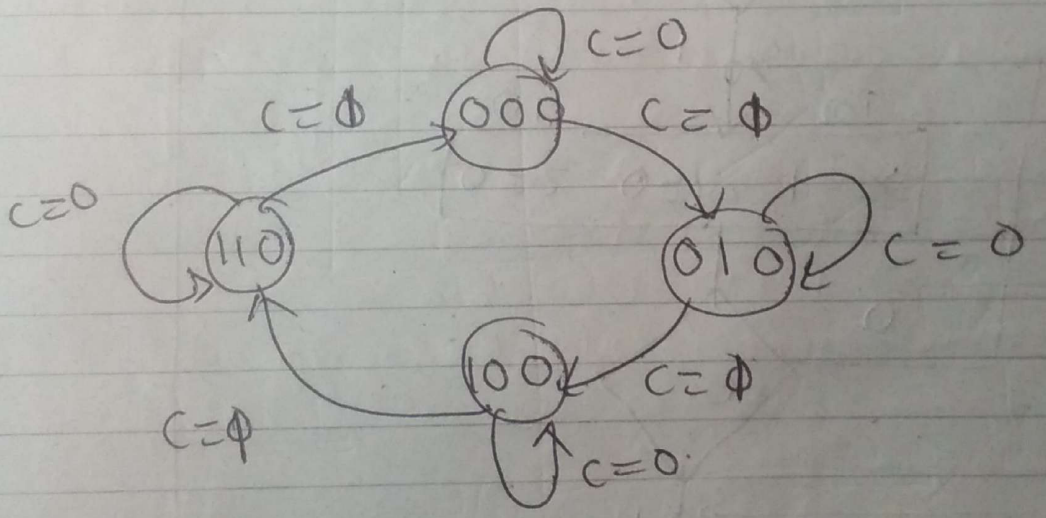
ii) If C=0 then ckt. hold present states.

⇒

Step ① state diagram.

$q_0 = 000$ $q_1 = 010$

$q_2 = 100$ $q_3 = 110$

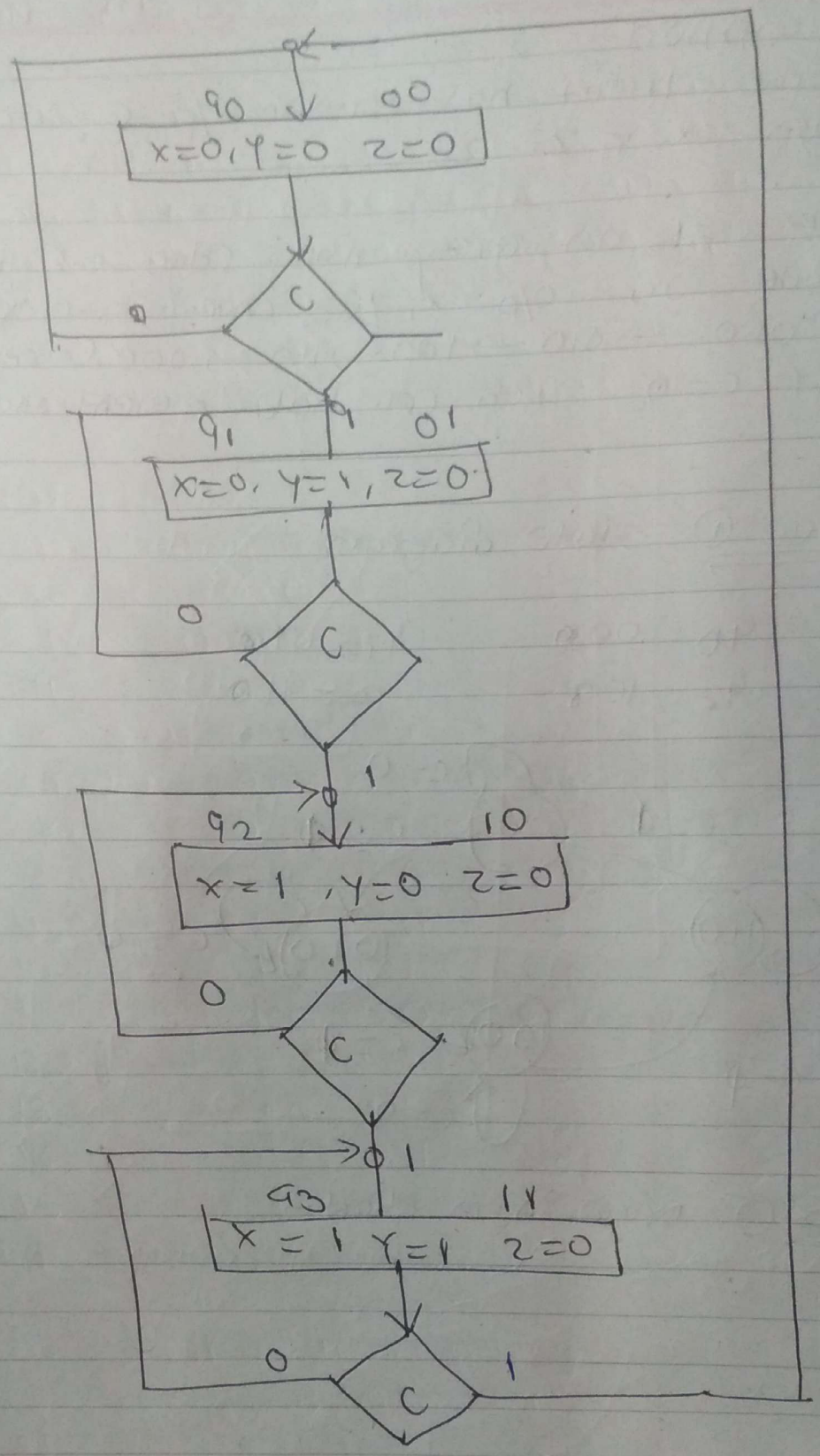


Step ② Draw ASM chart :-

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O/P X - Y - Z.



ASM Chart.

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④ Draw an ASM chart & state table for the synchronous ckt. having the following description.

"The ckt. has a control x', clock & o/p A & B. If x=1 on every clock rising edge the code on BA changes from 00 → 01 → 10 → 11 → 00 and repeat 'If x=0 the ckt holds the present states.

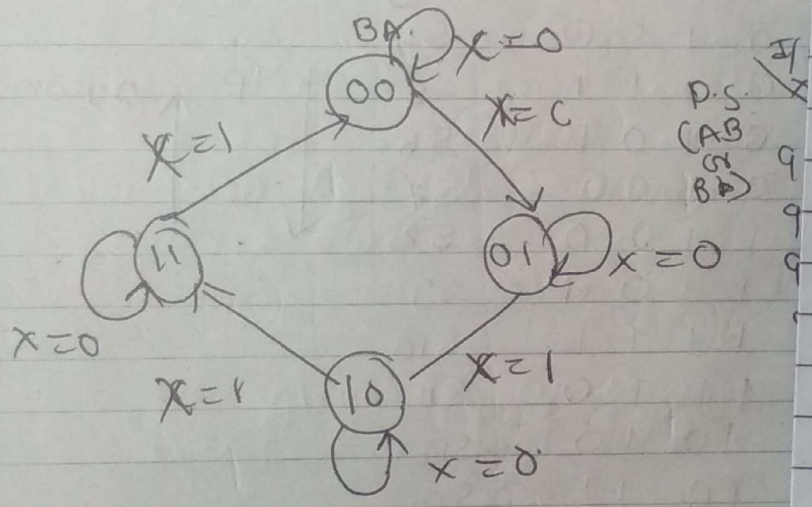
⇒ $q_0 = 00$ represented by

$q_1 = 01$

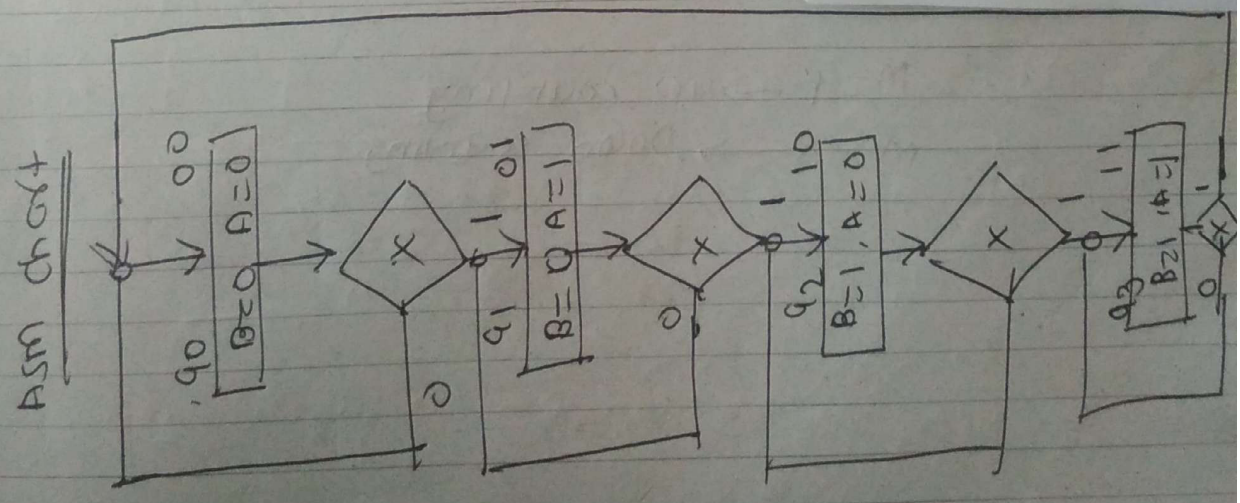
$q_2 = 10$

$q_3 = 11$

Step ① :- Draw state diagram.



④ State diagram.


ASM chart
00
01
B=0
A=0
B=1
x
x
x
x
0
0
0

Q 3 Q) Draw the ASM chart for 3-bit down counter.

3-bit down counter

E=0 counting Enable

E=1 counting disable

Also draw state diagram.

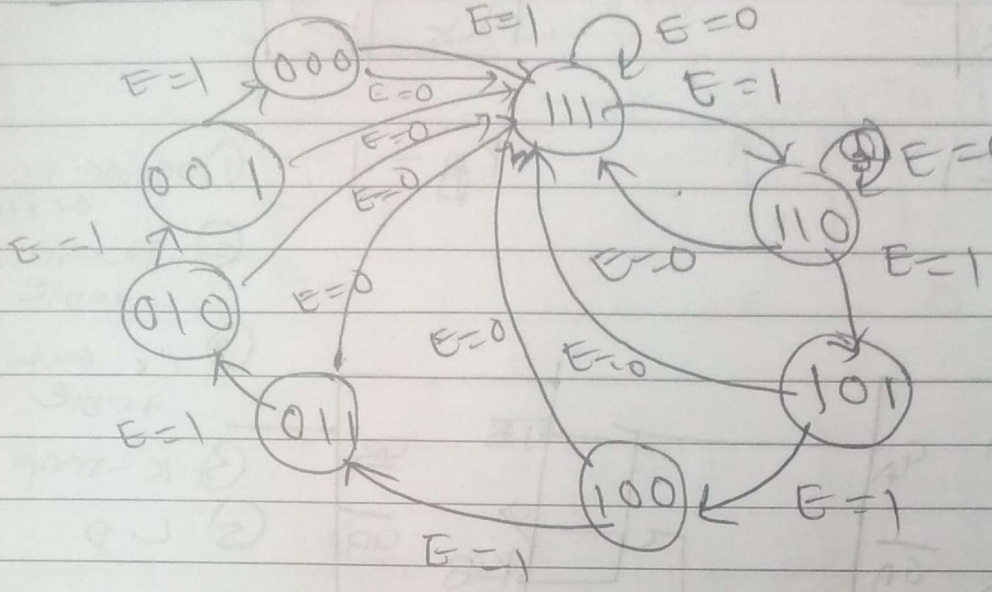
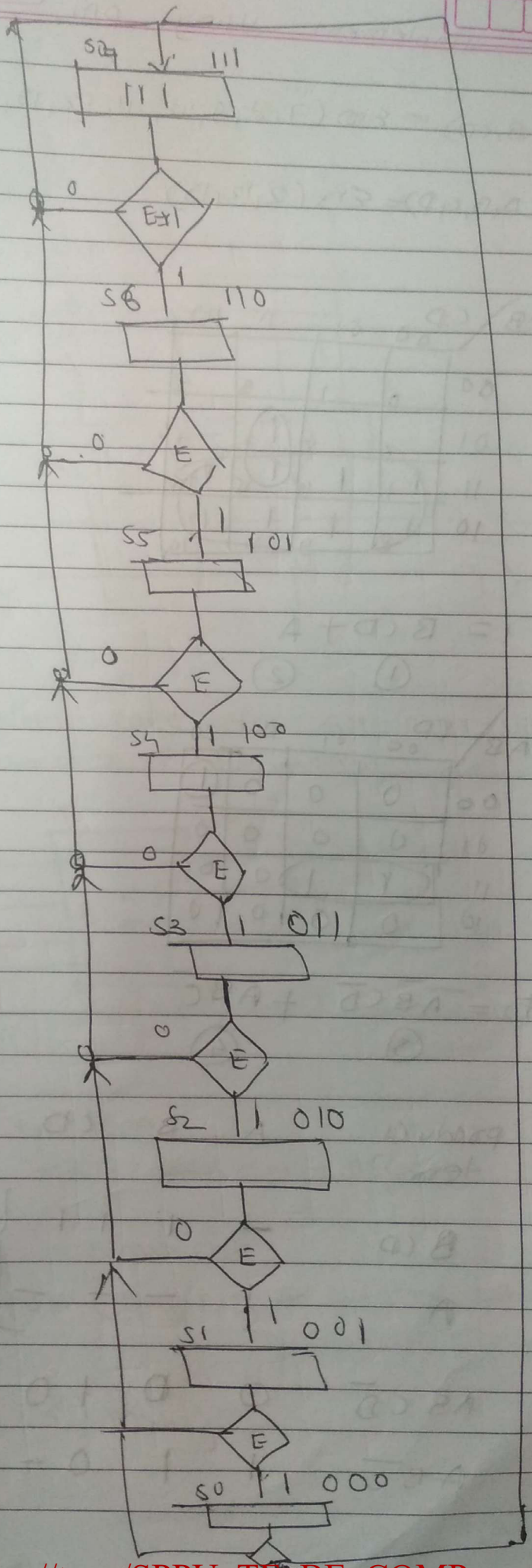


Fig:- State diagram

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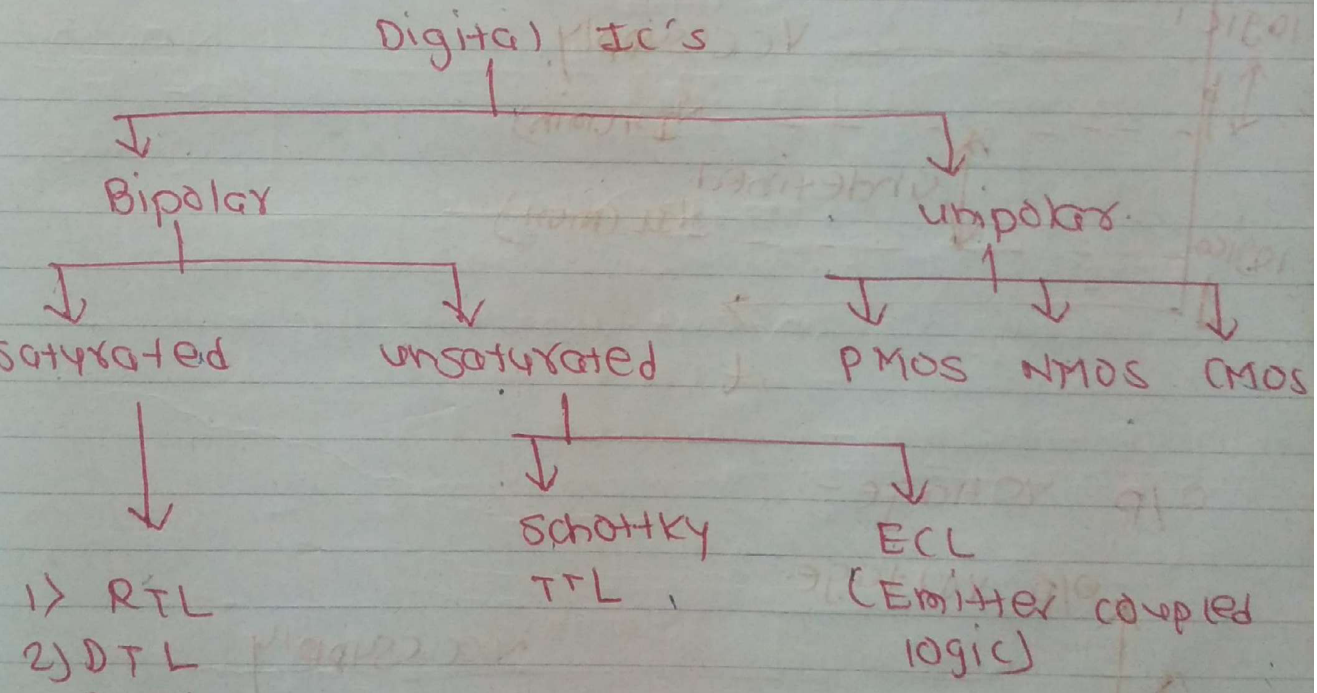
Logic families

Classification of logic families:-

1) Explain standard TTL characteristics in brief.
[6 Marks] (Dec-2009)

Q → (1) What is logic family? Explain types of logic families in detail
(6 Marks) — May 2010

Classification of logic families



Two basic technique for manufacturing Ic's
1) Bipolar technology
2) MOS technology

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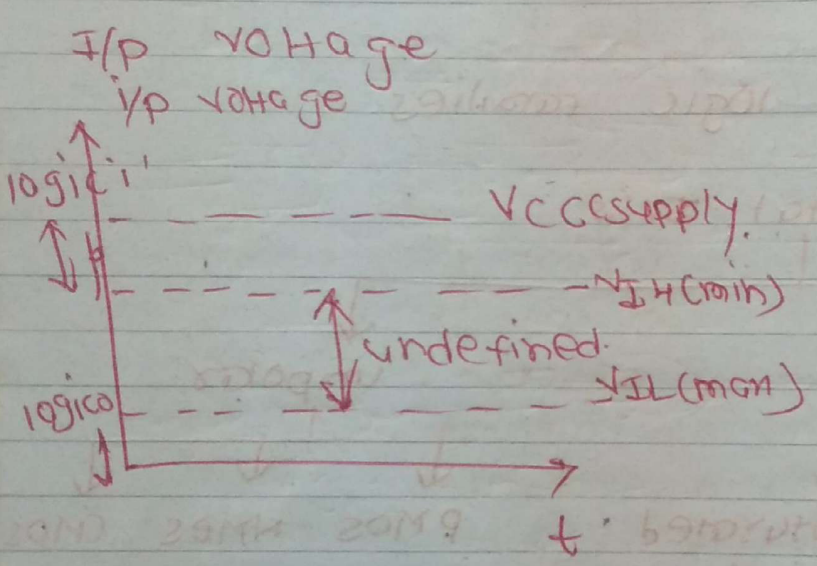
Digital IC parameters (characteristics) –:

voltage & current parameters –:

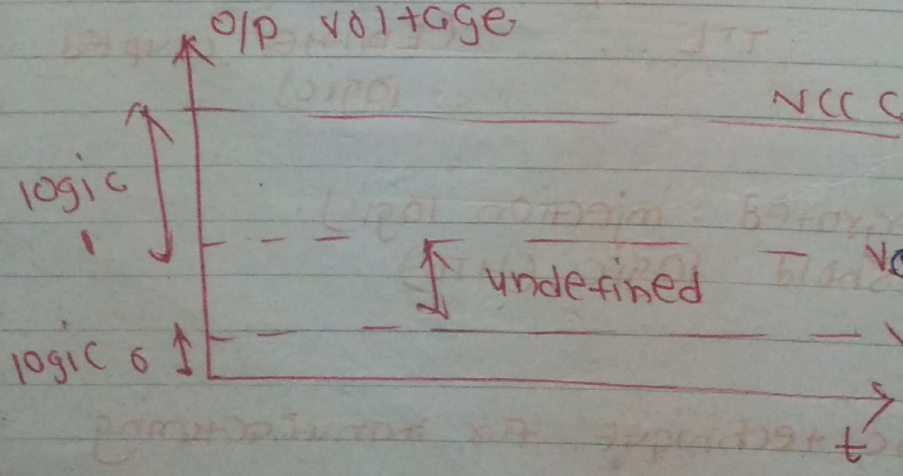
voltage parameters (threshold level) –:

1) V_{IL}(max)	– worst case	low level	i/p voltage
2) V_{IH}(min)	– Lower	High	
3) V_{OH}(min)	Lower	low	o/p
4) V_{OL}(max)	Higher	High	o/p voltage

voltage parameters:



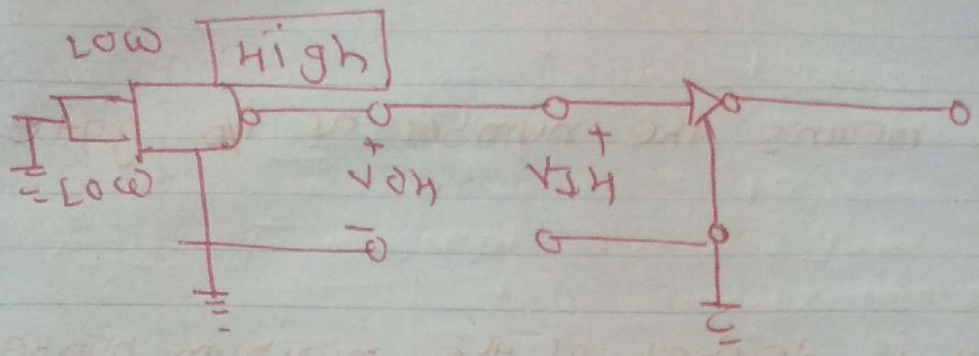
O/p voltage –:

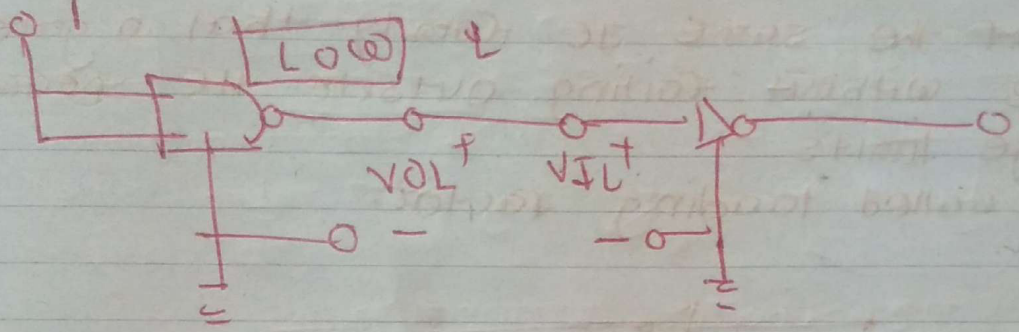


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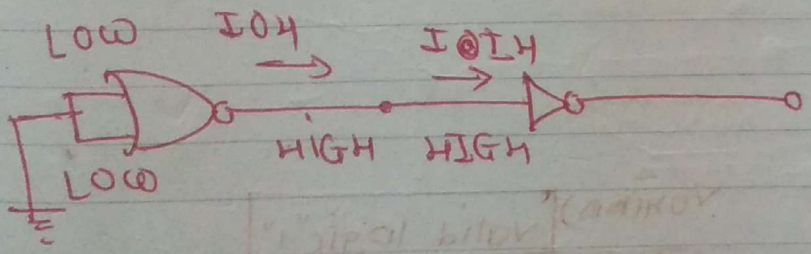
voltage parameter on a logic ckt.

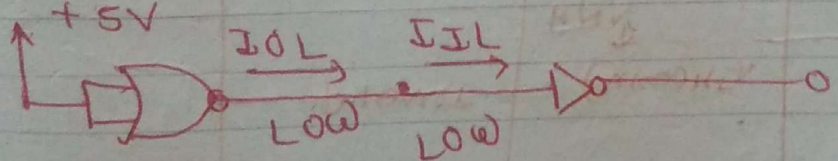




current parameters :-

1) I_{IL} :- Low level i/p current.
2) I_{IH} - High level i/p current.
3) I_{OL} - Low level o/p current.
4) I_{OH} - High level o/p current





current parameter

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Fan-in & Fan out :-

Fan in :-

Fan in means the number of i/p gate has.

Fan out :-

Fan out is defined as the maximum number of i/p of the same IC family that a gate can drive without failing outside the specified o/p voltage limits.

Also called loading factor.

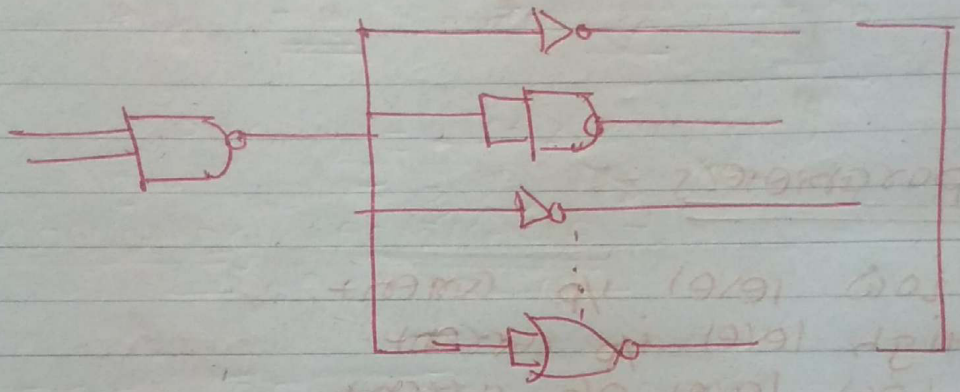
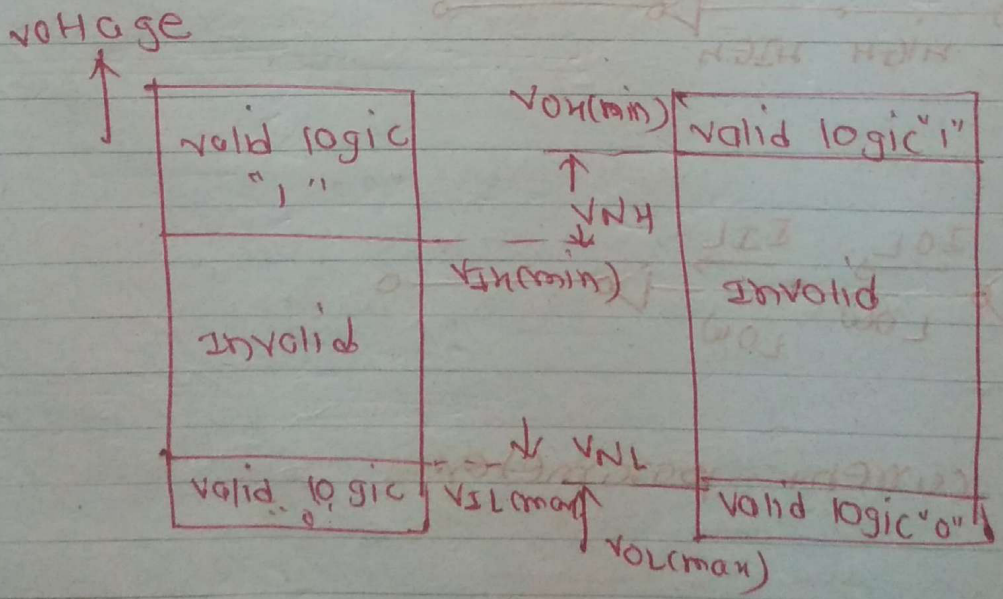


fig :- Fan-out.

Noise Margin :-



(a) I/p profile (b) o/p profile.

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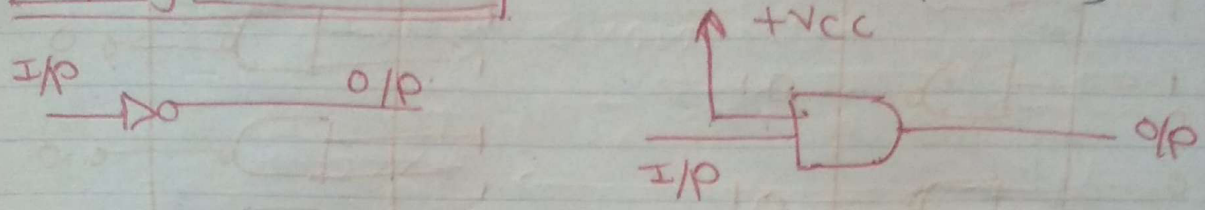
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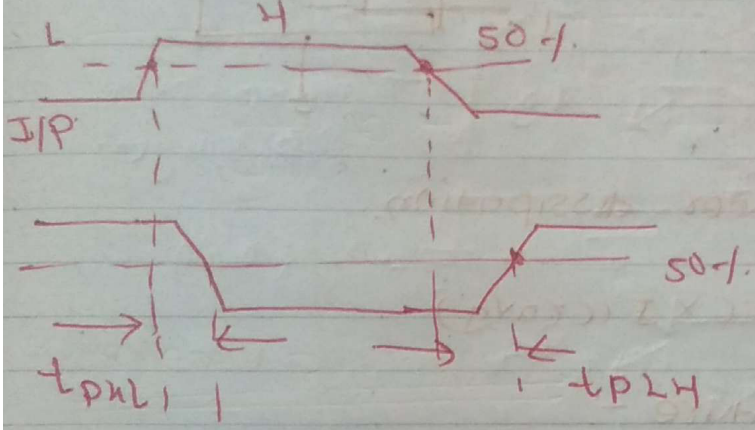
High level noise margin

$V_{NH} = V_{OH(min)} - V_{IH(min)}$

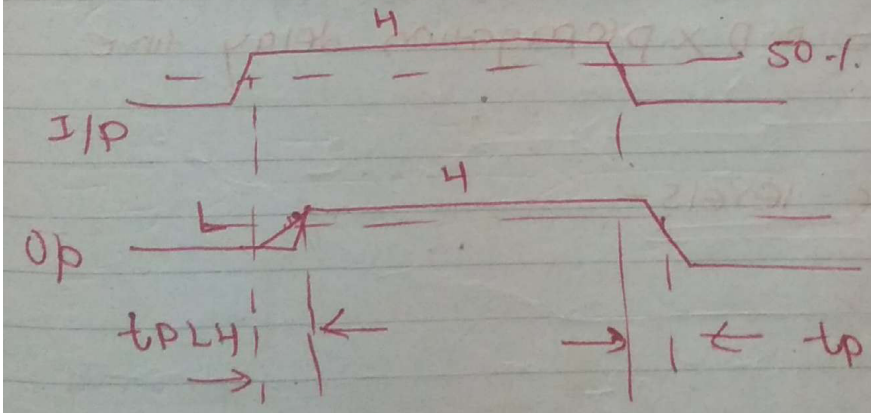
$V_{NL} = V_{IL(max)} - V_{OL(max)}$

Propagation delay :- (Speed of operation)





(a) propagation delay for an inverter.



(b) propagation delay for an AND gate

Power dissipation :- unit = mW

$P = V_{CC} \times I_{CC}$

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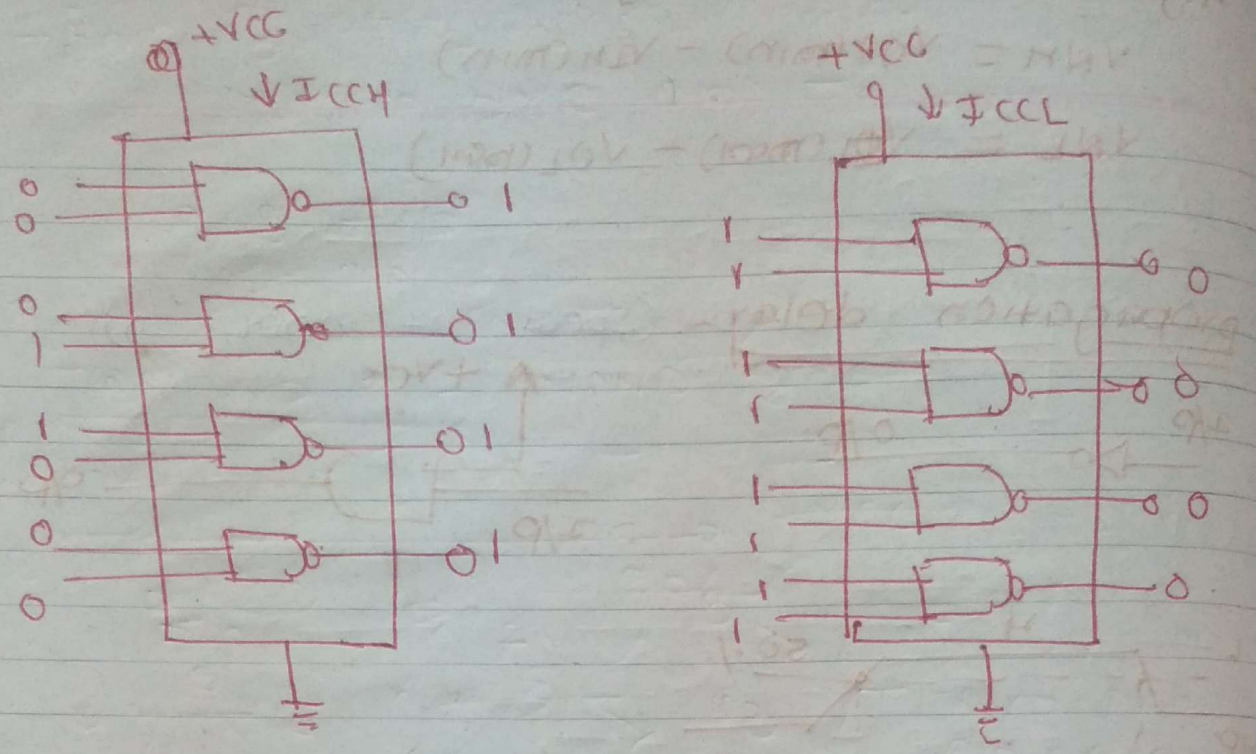


fig :- power dissipation.

$P_{D(avg)} = V_{CC} \times I_{CC(avg)}$

Operating temprature :-
— 0°C to 70°C.

Figure of merit :- (low)

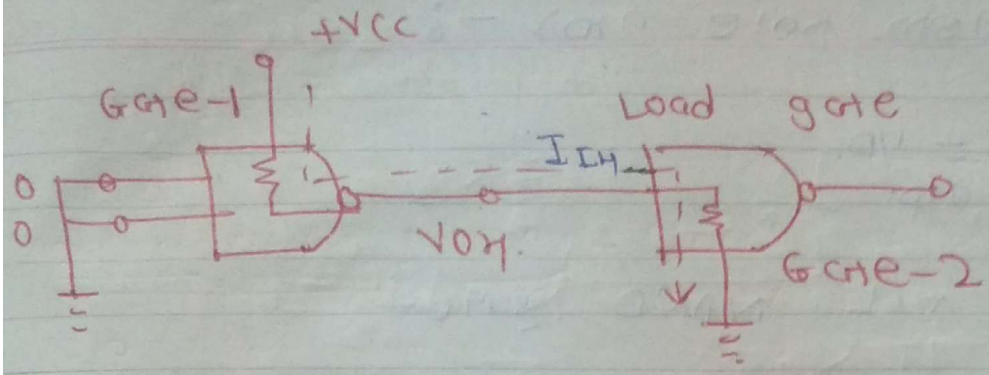
Figure of merit = P.D x propagation delay time

Invalid voltage levels :-

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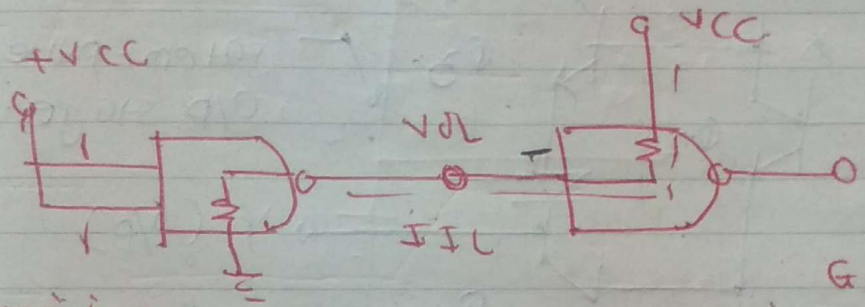
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current sourcing & current sinking.



Gate-1 sources current to gate-2

(a) current sourcing.



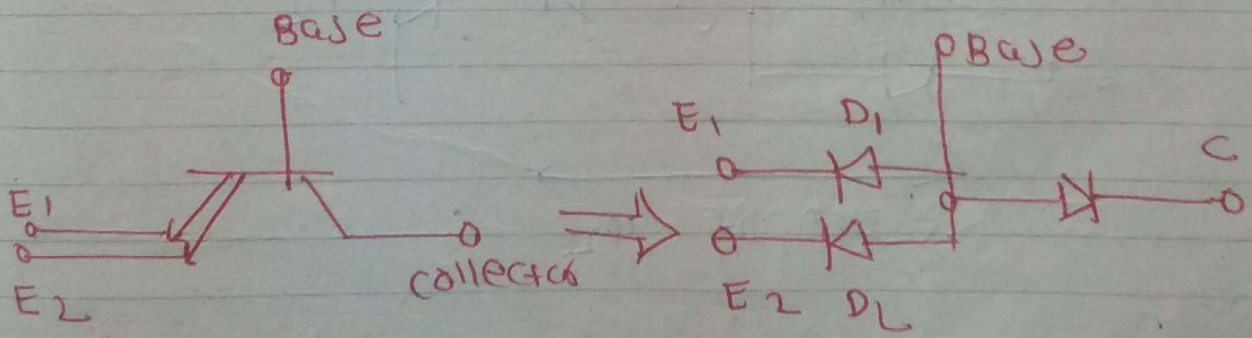
Driving gate

Gate-1 receives (sinks) current from gate-2

(b) current sinking.

* 7400 TTL series:-

The multiple emitter transistor:-



(a) multiple Emitter transistor.

(b) Eqn ckt.

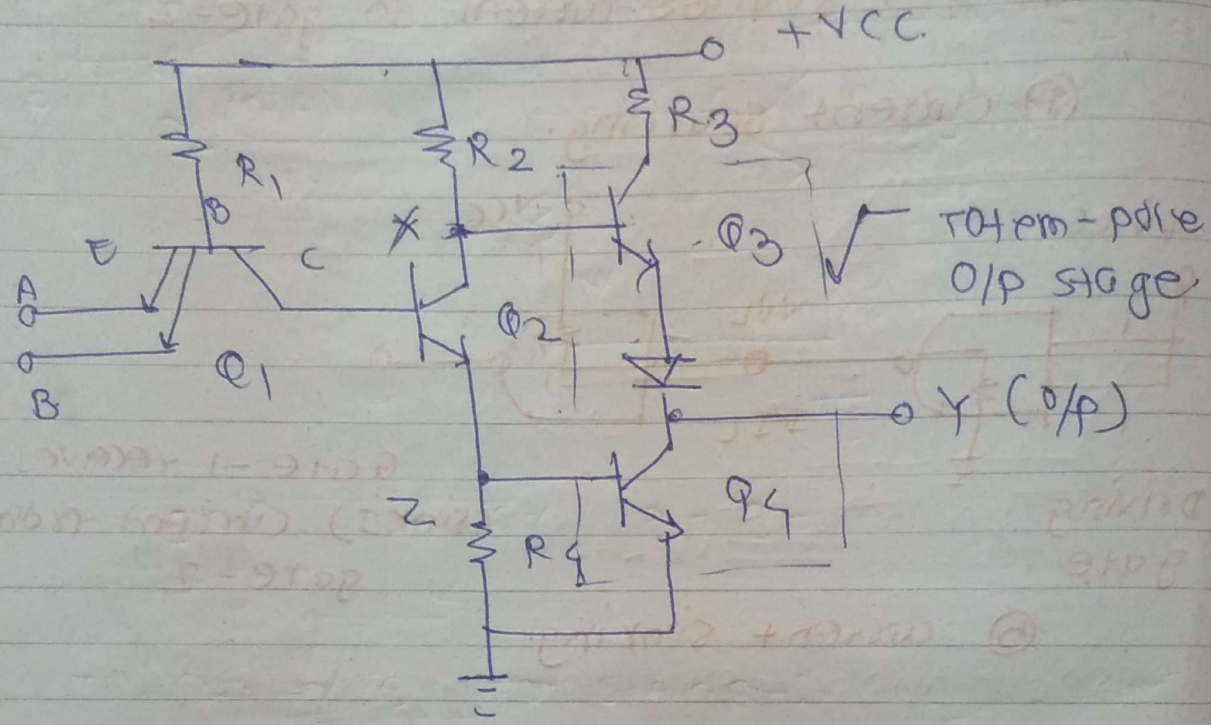
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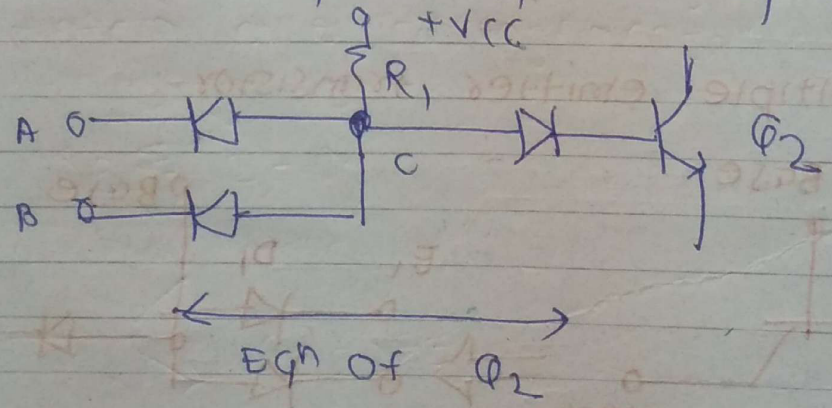
* TWO i/p TTL NAND gate:
(Totem pole O/P) :-

A, B = i/p
Y = O/p

TWO i/p TTL NAND gate-



Transistor Q_1 is replaced by equivalent.



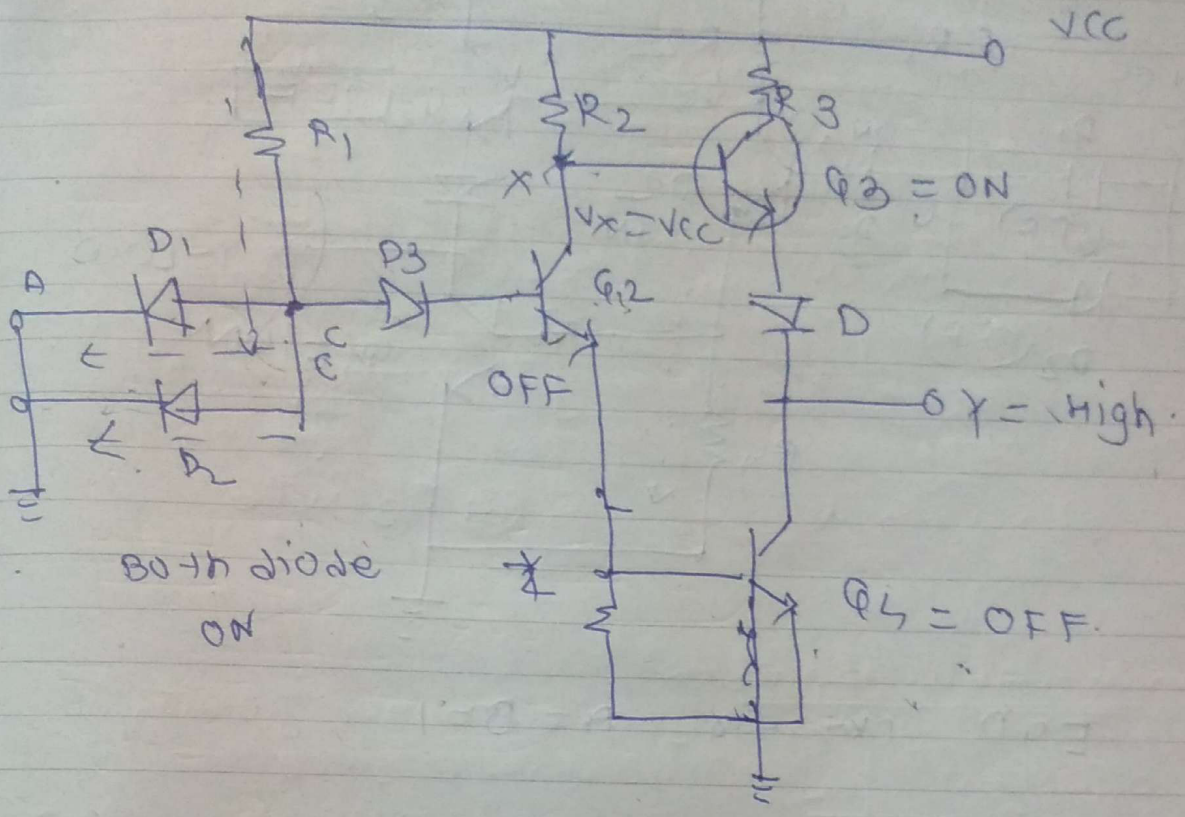
I) A & B both low
1) A & B connected to ground
2) Diode D_1 & D_2 start conducting
3) B-E of Q_1 is forward bias.
Due to voltage insufficient Q_2 is OFF

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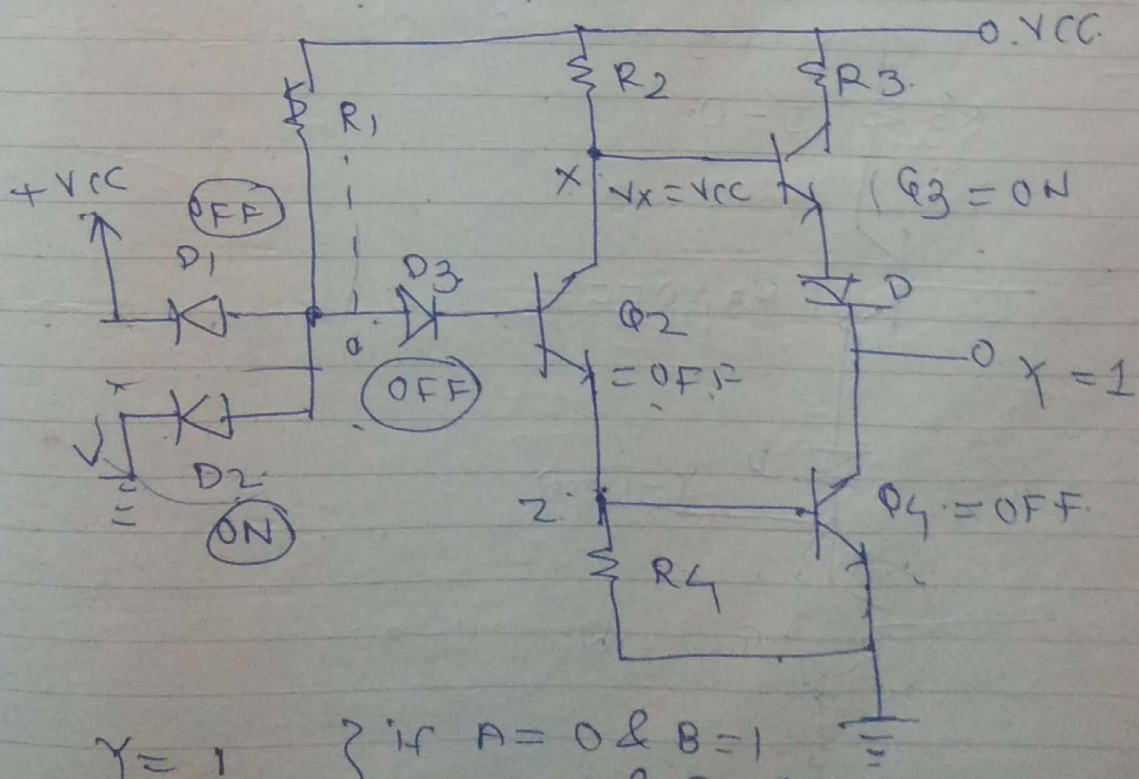
Transistor Q3 is operating in emitter - follower mode. o/p will be pulled to high voltage.

Y = 1 (High) — A = B = 0 (Low).



(a) EQN ckt. for A = B = 0

II) Either A or B low -:

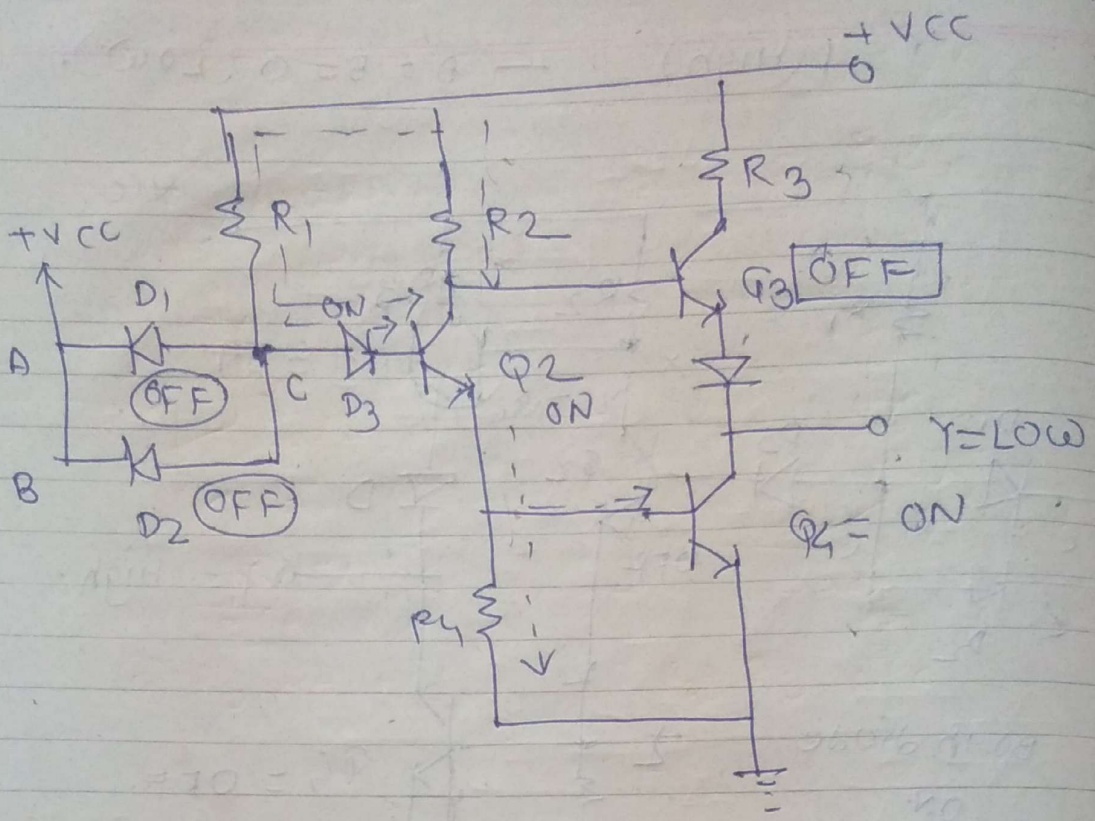


Y = 1 } if A = 0 & B = 1
& B = 0

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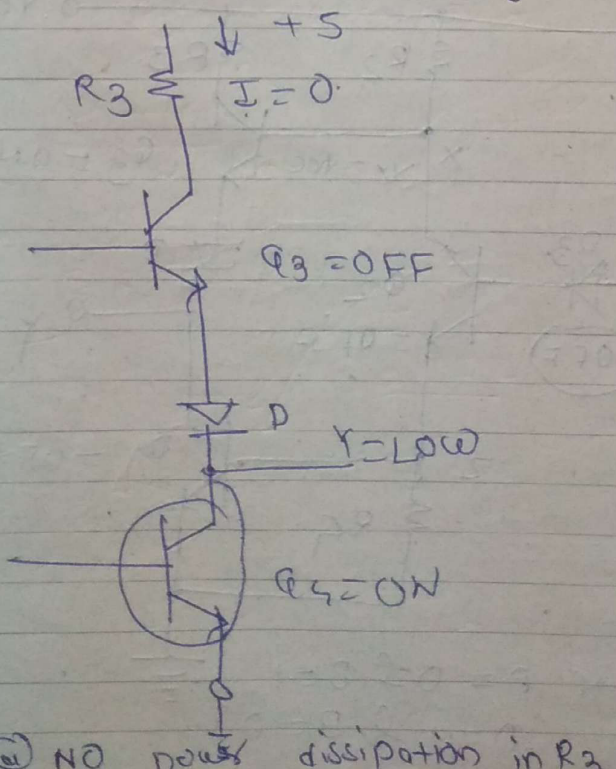
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A & B both High :- A=B=1 7400 ××
5400 ××



Eqn ckt. for A=B=1

Totem pole o/p stage :-
The arrangement of Q_3 & Q_4 on the o/p side of a TTL NAND gate is called as totem pole arrangement.



(a) NO Power dissipation in R_3

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① Open collector TTL :-

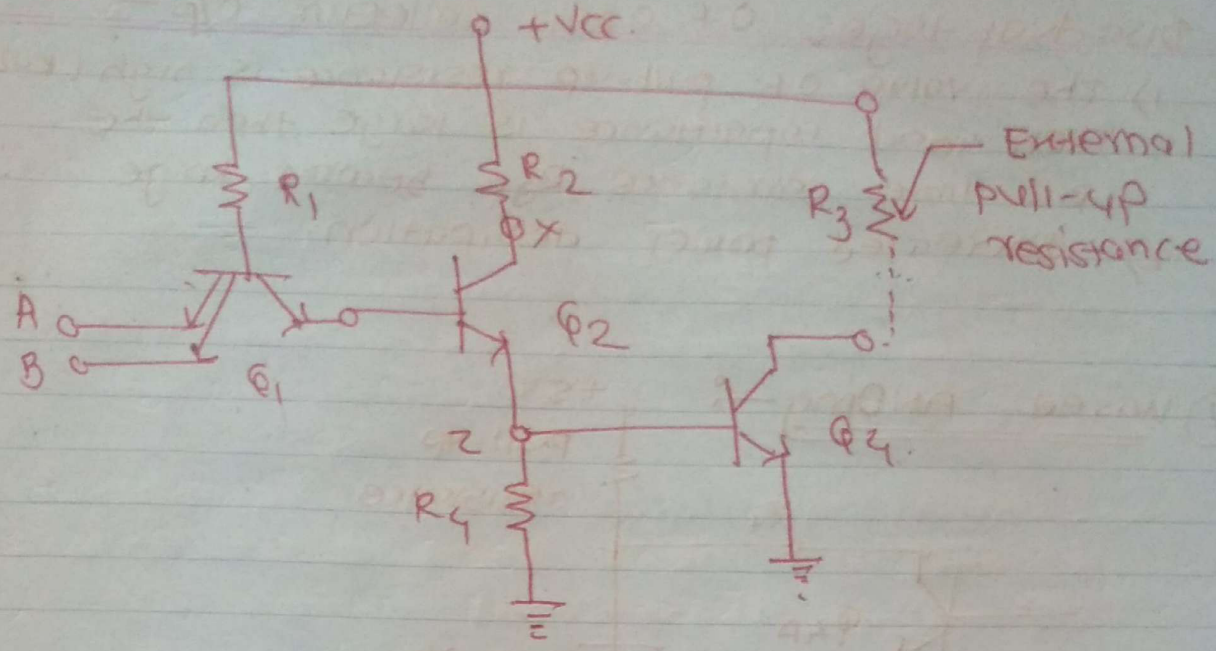


fig :- open collector 2 i/p NAND gate

Operation :-

I) with A = B = 0

Q_1 = Forward biased Q_1 = ON

Q_2 = OFF. Q_4 = OFF.

Collector voltage equal to Vcc. $\therefore Y = 1$ when A = B = 0

II) with A = 0, B = 1 OR A = 1, B = 0

Q_1 = ON $Q_2 = Q_4$ = OFF.

collector voltage = Vcc

$\therefore Y = 1$

III) with A = B = 1

Q_1 = OFF Q_2 = ON

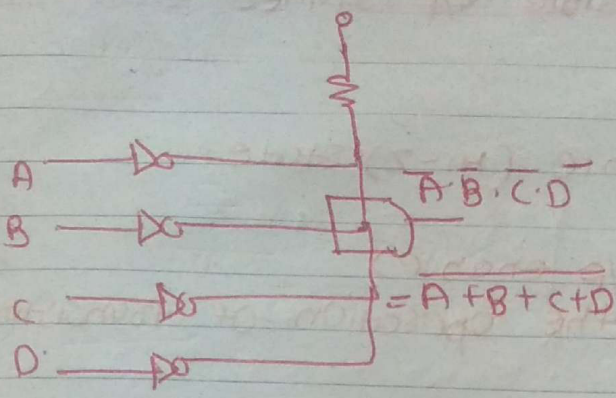
Q_4 = ON

$Y = 0$

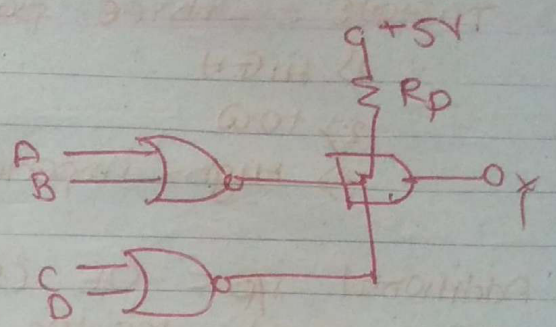
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o/p (c) wired ANDing of inverter o/p.



(d) wired ANDing of the NOR o/p.



$Y = (\overline{A+B}) \cdot (\overline{C+D})$

$= \overline{A+B+C+D}$

(III) comparision of totem-pole & open collector o/p.

Sr. No.	parameter	Totem-pole	open-collector.
1)	circuit component on o/p side	Q_3 (pull-up transistor) D & Q_4 (pull-down) transistor used.	only the pull down transistor Q_4 is used.
2)	wired ANDing	cannot be done	Easily be done.
3)	External pull-up resistor.	NO required	required to be connected.
4)	speed	Operating speed is high	Operating speed is low.
5)	power dissipation.	Low due to presence of pull-up transistor Q_3	High due to current flowing through R_3

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* 74C00 CMOS

1) CMOS Inverter
2) CMOS NOR
3) CMOS NAND

NMOS = ON i/p = 1
= OFF i/p = 0

PMOS = ON i/p = 0
= OFF i/p = 1

1) CMOS Inverter :-

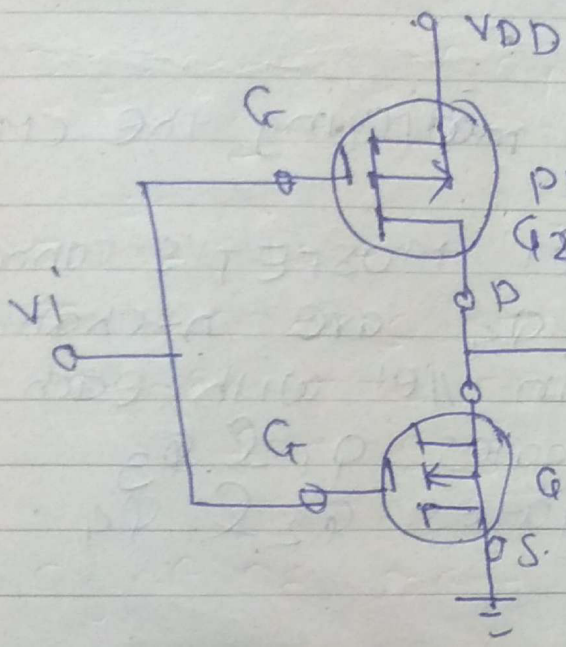


fig-i
CMOS inverter with +ve voltage

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Operation

I) v_i = 0 VOHS (logic 0)

Q_2 = ON & Q_1 = OFF.

Y = 1

II) v_i = VDD (logic 1)

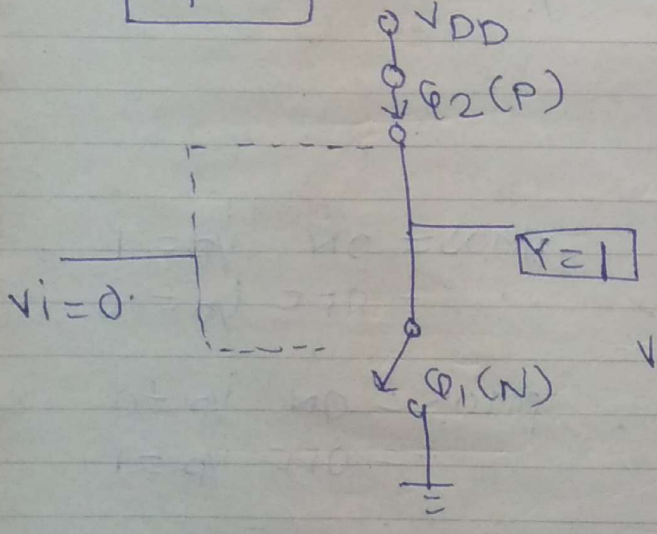
Q_1 = ON Q_2 = OFF

o/p will be connected to GND.

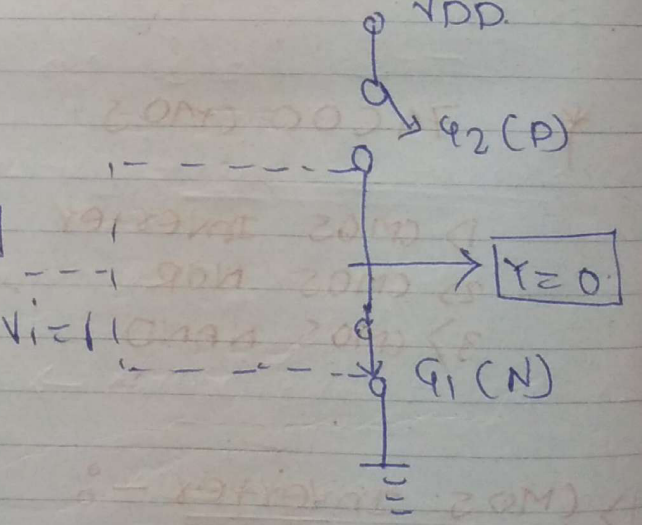
Y = 0

Eqn ckt :-

① $V_i = 0$



② $V_i = 1$



2) CMOS NOR gate

1) It is obtained by modifying the CMOS inverter circuit.
2) Q_1 & Q_2 are p-channel MOSFET'S connected in series and Q_3 & Q_4 are n-channel MOSFET'S connected in ||el with each other
3) I/p A connected to gates Q_1 & Q_3
   B connected to gates Q_2 & Q_4.

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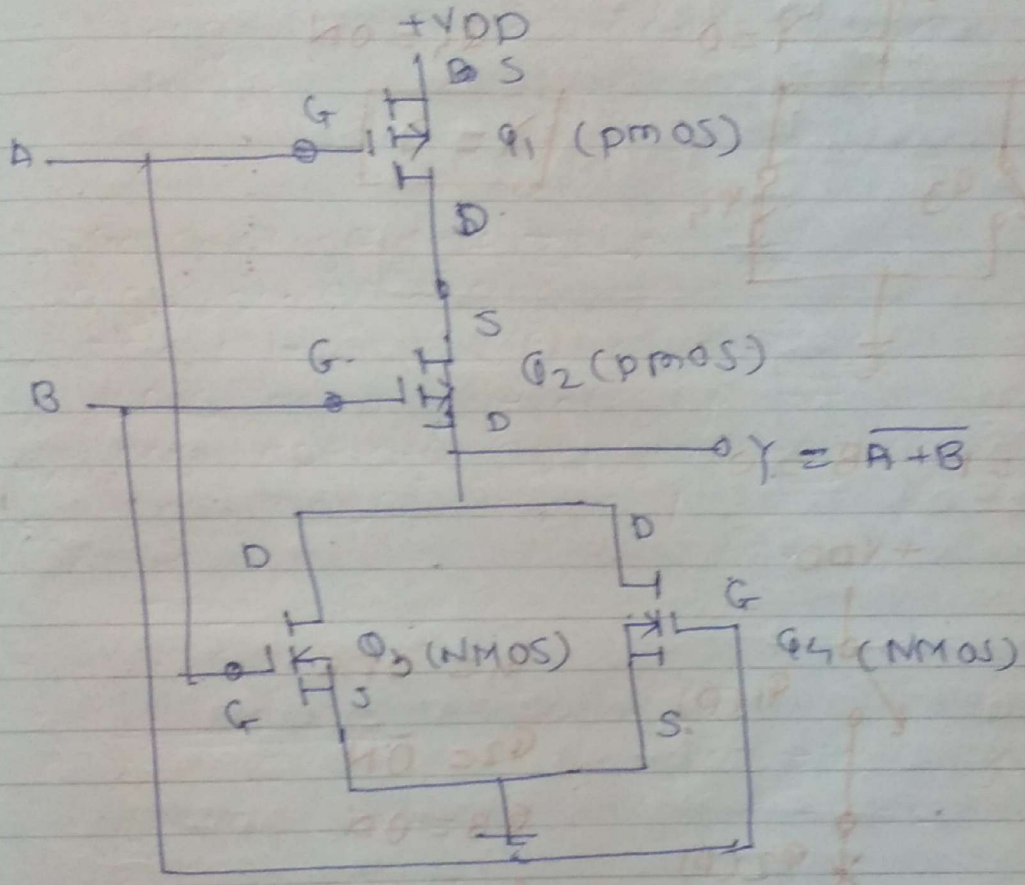
O/p

Y = 1 if $Q_1 = Q_2 = ON$ & $Q_3 Q_4 = OFF$

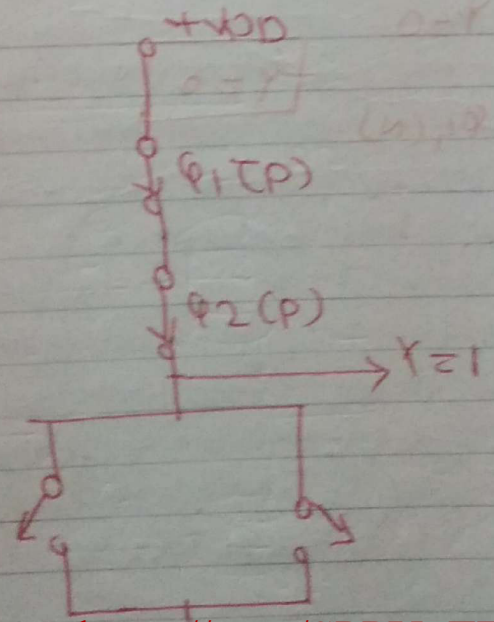
Y = 0 if $Q_1 = Q_2 = OFF$ & $Q_3 Q_4 = ON$

Operation :-

1) with A = B = 0



⇒



Q_1 = ON

Q_2 = ON

Q_3 = OFF

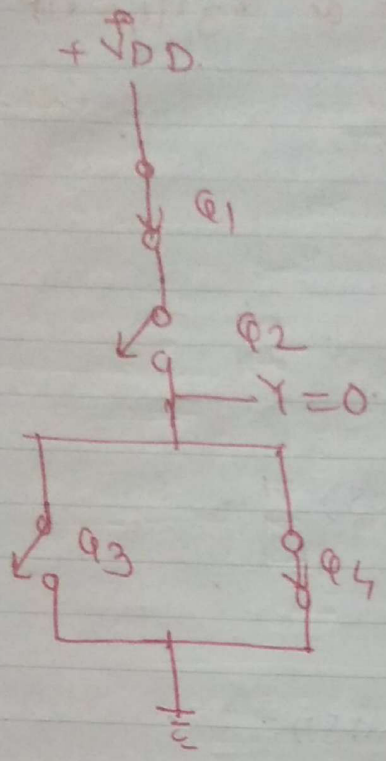
Q_4 = OFF

Y = 1

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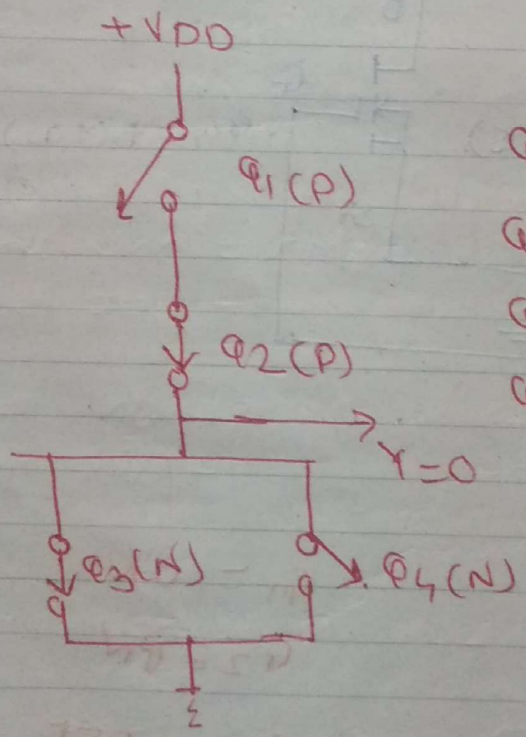
A = 0 & B = 1



Q_1 = ON
Q_2 = OFF
Q_3 = OFF
Q_4 = ON

Y = 0

③ A = 1 B = 0



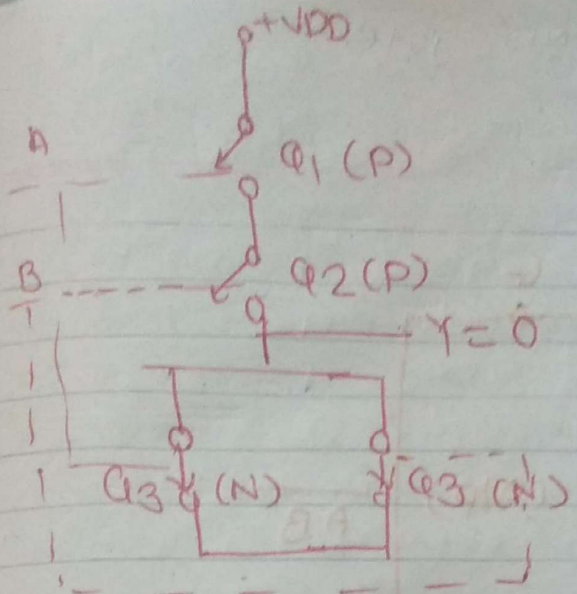
Q_1 = OFF
Q_2 = ON
Q_3 = ON
Q_4 = OFF

Y = 0

④ A = 1 , B = 1

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$Q_1 = Q_2 = OFF$

$Q_3 = Q_4 = ON$

$Y = 0$

I/p		Transistors				O/p (Y)
A	B	Q_1	Q_2	Q_3	Q_4	~~VDD(1)~~
0	0	ON	ON	OFF	OFF	VDD(1)
0	1	ON	OFF	OFF	ON	0
1	0	OFF	ON	ON	OFF	0
1	1	OFF	OFF	ON	ON	0

3) CMOS NAND gate :-

Q_1 & Q_2 are connected in parallel (//el).
(P-channel)

Q_3 & Q_4 are connected in series.
↳ (n-channel)

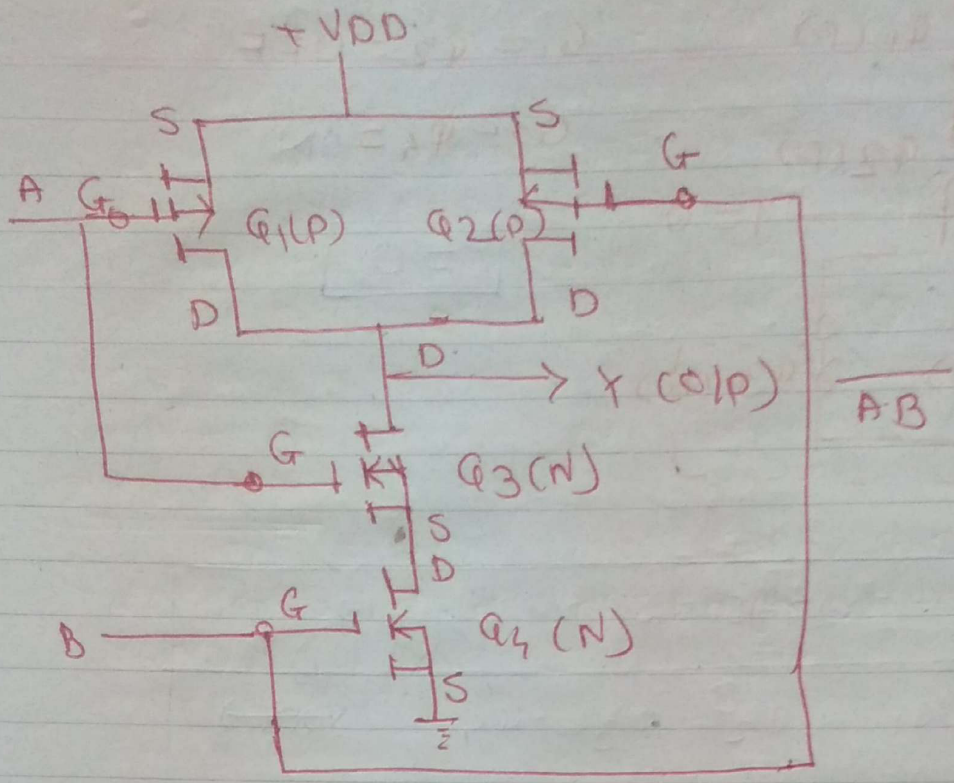
1) I/p A connected to gates Q_1 & Q_3 so control the status of MOSFET Q_1 & Q_3

2) I/p B connects to gate Q_2 & Q_4 so control status of MOSFET Q_2 & Q_4.

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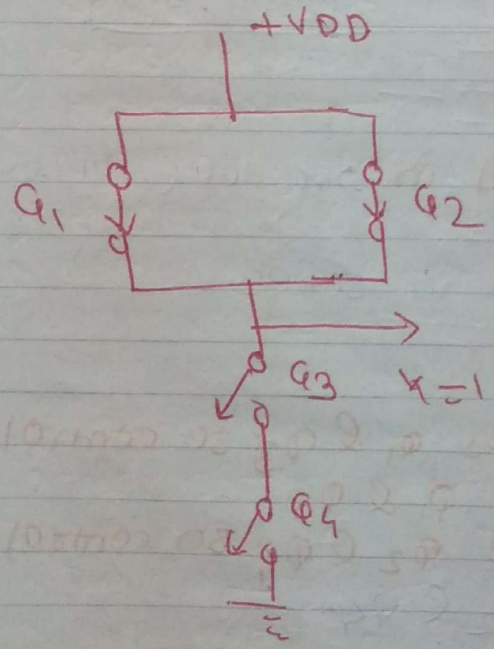
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fig :- 2 i/p CMOS NAND gate



Operation :-

1) A = B = 0



Q1 = Q2 = ON

Q3 = Q4 = OFF

Y = 1

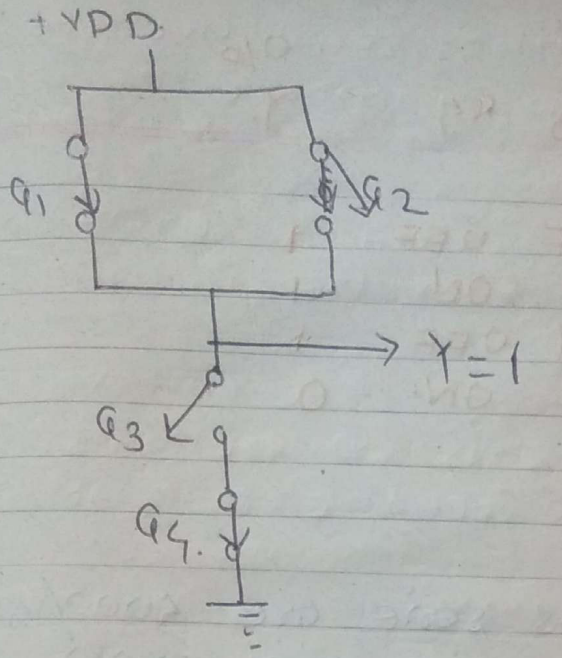
(2) A = 0, B = 1

Q1 = ON, Q2 = OFF

Q3 = OFF Q4 = ON

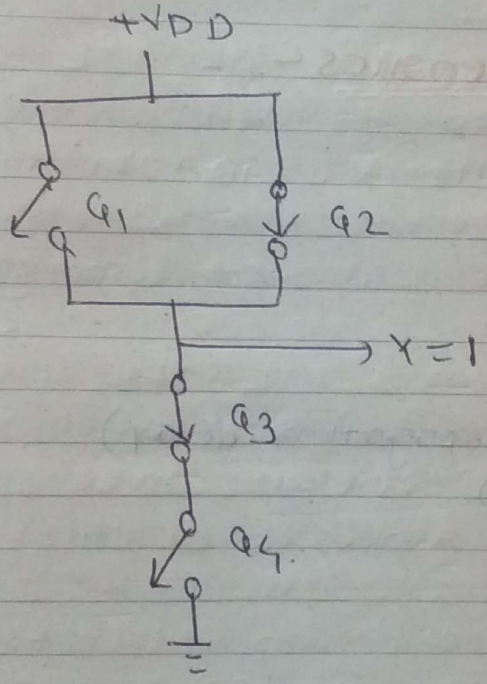
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Y=1

③ A=1 B=0

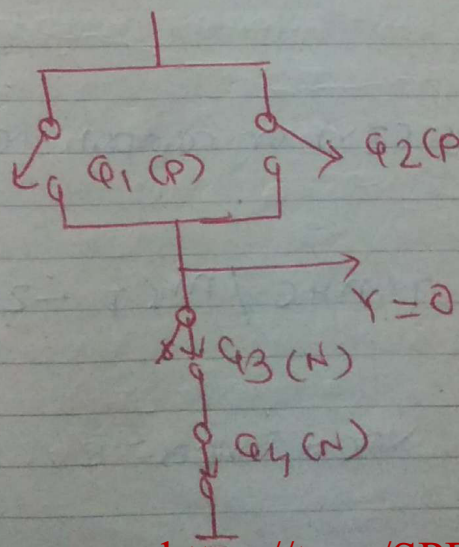


Q1 = OFF Q2 = ON

Q3 = ON Q4 = OFF.

Y=1

④ +VDD A=1 B=1



Q1 = Q2 =

Q3 = Q4 =

Y=0

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f

I/P		Transistor				O/P
A	B	Q_1	Q_2	Q_3	Q_4	Y
0	0	ON	ON	OFF	OFF	1
0	1	ON	OFF	OFF	ON	1
1	0	OFF	ON	ON	OFF	1
1	1	OFF	OFF	ON	ON	0

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* Comparision of CMOS & TTL

Sr. No.	parameter	CMOS.	TTL.
1>	Device used	N-channel & p-channel MOSFET.	BJT.
2>	V_{IH}(min)	3.5 V	2 V
3>	V_{IL}(max)	1.35 V	0.8 V
4>	V_{OH}(min)	4.95 V	2.4 V
5>	V_{OL}(max)	0.05 V	0.4 V.

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* compare TTL & CMOS logic family. Also draw CMOS NOR gate - (Dec-2009)

6)	V_{NH}	$V_{NH} = 1.45V$	0.4V
7)	V_{NL}	$V_{NL} = 1.45V$.	0.4V
8)	Noise Immunity	Better than TTL.	Less than CMOS.
9)	propagation delay	10 ns (metal gate CMOS)	10 ns.
10)	S/W speed	less than TTL	Faster than CMOS.
11)	Fan-out	50	10
12)	speed-power product	10 SPJ	100 pJ
13)	Dependance of P.D on freq^n	P.D increases with increase in freq^n	PD does not depend on freq^n.
14)	power dissipation per gate	PD = 0.1 mW	10 mW.
15)	power supply voltage	3V to 15V	fixed equal to 5V.
16)	unconnected i/p	Unused i/p connected to GND or V_{DD}. They should never be left floating.	I/p can remain floating. The floating i/p are treated as logic '1'.
17)	operating areas.	MOSFET are operated as S/W i.e. in the ohmic region or CUT-off region.	transistor are operated in saturation & cut-off region.

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